

TPG4190 Seismic data acquisition and processing

Lecture 2: Receivers

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Overview

1. Receiver system (sec. 2.8)

Receiver system

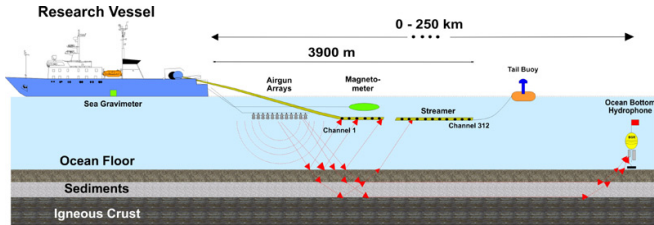


Figure: Marine seismic setup

- ▶ Each channel is a group of hydrophones (10-20) connected together
- ▶ Distance between each group is typical 12.5m or 6.25m
- ▶ Hydrophones are pizzo-electric devices sensitive to pressure

Receiver system

A single group of hydrophones are summed together as

$$\sum_{j=0}^{N-1} a(x_j), \quad (1)$$

where $x_j = j\Delta x$ and Δx is the distance between hydrophones and N is the number, while $a(x_j)$ is the amplitude measured at position x_j . Formally, this is the same as

$$g(x) = \sum_{j=0}^{N-1} \int_0^{x_j} dx \delta(x_j - x) a(x) \quad (2)$$

The expression inside the sum can be recognized as the convolution of $a(x)$ and $\delta(x_j - x)$. The Fourier transform of the convolution of two functions is equal to the product of the Fourier transform of each function. Hence we have

$$G(k) = A(k) \sum_{j=0}^{N-1} \Delta(k, x_j) \quad (3)$$

Receiver system

where Δ is the Fourier transform of the delta function

$$\Delta(k) = \exp(ikx_j) = \exp(ikj\Delta x) \quad (4)$$

Hence we have for the Fourier transform of the sum of the hydrophones

$$H(k) = \sum_{j=0}^{N-1} \exp(ikj\Delta x). \quad (5)$$

This is a geometric series and the sum is equal to

$$\sum_{j=0}^{N-1} \exp(ikj\Delta x) = \frac{1 - \exp(ikN\Delta x)}{1 - \exp(ik\Delta x)}. \quad (6)$$

The right hand side is rewritten as

$$\frac{\exp(ikN\Delta x/2) [\exp(-ikN\Delta x/2) - \exp(ikN\Delta x/2)]}{\exp(ik\Delta x/2) [\exp(-ik\Delta x/2) - \exp(ik\Delta x/2)]}. \quad (7)$$

Receiver system

which is recognized as:

$$H(k) = \frac{\sin(kN\Delta x/2)}{\sin(k\Delta x/2)} \quad (8)$$

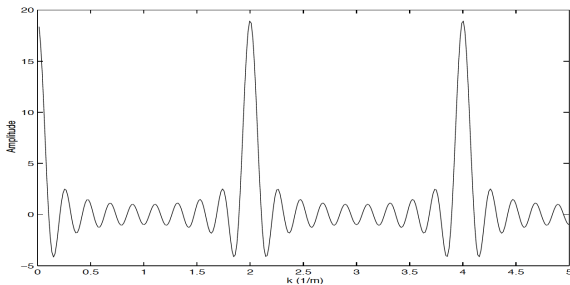


Figure: Receiver group array spectrum

Receiver system

- ▶ Dynamic range of seismic data: $1\mu\text{bar}$ - 1 bar
- ▶ Seismic data often low-pass filtered at 125Hz
- ▶ Old data often High-pass filtered at 3-5Hz
- ▶ Temporal aliasing not a problem
- ▶ Spatial aliasing almost always problematic

Plane waves

A simple model of seismic data is a plane wave

$$\exp(i\omega t - ik_x x - ik_z z), \quad (9)$$

where the wavenumber vector $\mathbf{k} = (k_x, k_z)$

$$k_x = \frac{\omega}{v} \sin(\theta), \quad (10)$$

$$k_z = \frac{\omega}{v} \cos(\theta). \quad (11)$$

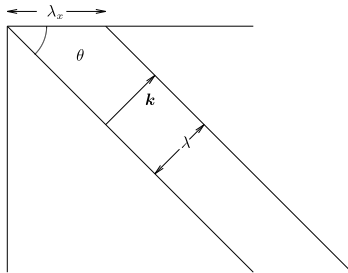
θ is the angle with the vertical.

The frequency ω and wave velocity v is Here related to the the component of the wavenumber vector by

$$\frac{\omega^2}{v^2} = k_x^2 + k_z^2, \quad (12)$$

which is the dispersion relation for plane waves.

Plane waves



Sampling of wavefields

Fourier transform over the horizontal coordinate x , relating the spatial domain with the wavenumber domain:

$$p(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} dk_x P(k_x) \exp(-ik_x x), \quad (13)$$

$$P(k_x) = \int_{-\infty}^{+\infty} dx p(x) \exp(ik_x x). \quad (14)$$

Sampling of wavefields

In chapter 2 we derived the Nyquist sampling criterion

$$\Delta t \leq \frac{1}{2f_{Nyq}}. \quad (15)$$

Expressed with the angular frequency the sampling criterion becomes

$$\Delta t \leq \frac{\pi}{\omega_{Nyq}}, \quad (16)$$

where $\omega_{Nyq} = 2\pi f_{Nyq}$.

$$\Delta x \leq \frac{\pi}{k_x^{Nyq}}, \quad (17)$$

where k_x^{Nyq} is the largest wavenumber. The last equation becomes

Sampling of wavefields

$$\mathbf{k}_x^{Nyq} \leq \frac{\pi}{\Delta x}. \quad (18)$$

By using equation (10), we get a relation between the angular frequency and the angle between the vertical and the propagation direction of the wave, often called the dip-angle

$$\frac{\omega}{v} \sin(\theta) \leq \frac{\pi}{\Delta x}. \quad (19)$$

Spatial aliasing

- ▶ Spatial aliasing: Spacing between receivers is too large to capture rapid variations of the data

$$\Delta r < \frac{v}{2f \sin(\theta)}. \quad (20)$$

where

- ▶ Δr : Spacing between receiver (groups)
- ▶ v : Velocity at the receiver
- ▶ θ : Angle between vertical and direction of wave

Receiver system

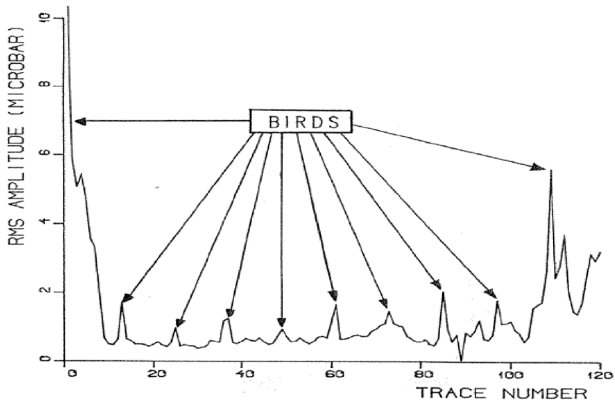


Figure: Streamer ambient noise

Receiver system

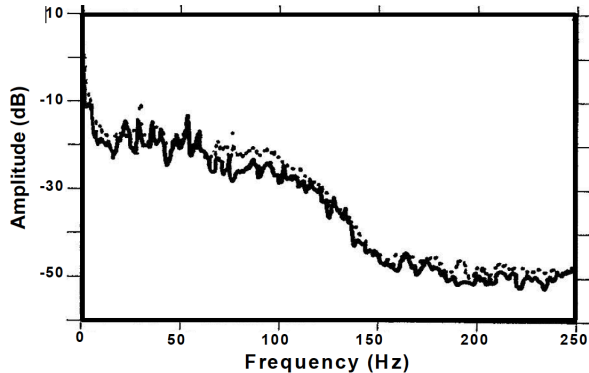


Figure: Noise record for two different sources

$$P_{db} = 20 \log \left(\frac{P}{P_{ref}} \right) \quad (21)$$

Receiver system

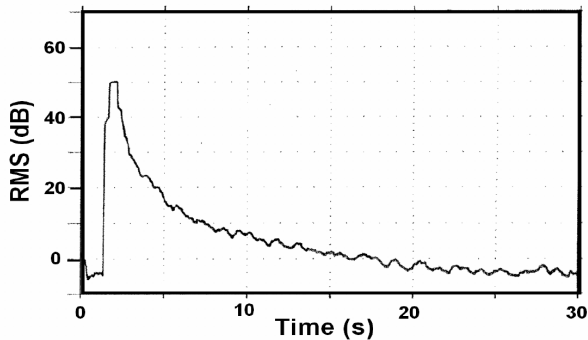


Figure: Noise record for single shot